BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 013 · Flowcharts, Pseudocode and Programming Constructs

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is abstraction?

Q2. Why might an HDD suit a large archive?
A. Low cost per GB
B. No moving parts
C. Unlimited lifespan
D. Always fastest access

Q3. Why is an SSD often faster than an HDD for random access?

Q4. A laptop is used on a bumpy bus. “Portability and durability mean the same thing.” Correct this. How should this affect the storage choice?

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

sequence — Executing instructions in a stated order.

selection — Choosing a branch based on a condition.

iteration — Repeating instructions.

decision diamond — Flowchart shape containing a condition with branches.

New learning

Sequence: Instructions run in order; arrows on a flowchart show control flow.

Selection: A decision tests a condition and follows one of the branches, often using IF/ELSE.

Iteration: Instructions repeat; a loop can be count-controlled or condition-controlled.

Flowchart symbols: Use terminators (start/end), parallelograms (input/output), rectangles (processes), diamonds (decisions) and arrows. Pseudocode expresses the logic as text.

Worked example

Design a pass/fail algorithm: mark 52 passes when mark >= 50.

1. Start by reading a mark.

2. Use a diamond decision: is mark >= 50?

3. True branch prints Pass; false branch prints Try again.

4. Both branches lead to End. Pseudocode uses IF, ELSE and ENDIF.

5. For mark 52, the true branch prints Pass.

We Do · Practise with guidance

Guided 1

Design a flowchart to check whether an age is at least 18.

Worked method: Input age; decision diamond age >= 18; adult/under 18 outputs; end.

Guided 2

Write pseudocode with IF/ELSE/ENDIF for the age-check example.

Worked method: IF age >= 18 THEN print('Adult') ELSE print('Under 18') ENDIF (valid equivalent accepted).

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Guided 3

Explain the difference between an IF statement and a FOR loop.

Worked method: An IF chooses a branch; a FOR repeats instructions a known number of times.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict how far the file exceeds available storage (GB).

```
fileGB = 12
freeGB = 10
print(fileGB - freeGB)
```

predict

Question 1

Name two flowchart symbols and their functions. [2 marks]

Question 2

Trace: x = 3; x = x + 4; print(x). Give the output and describe sequence. [3 marks]

Question 3

Explain the different purposes of selection and iteration in a password-length checker which asks again if too short. [4 marks]

Exit Ticket · Answer before checking

An IF statement automatically repeats until the condition changes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Removing irrelevant detail while keeping what is needed to solve the problem.

Q2. A — Low cost per GB

Q3. It does not need a moving head to seek positions on spinning platters.

Q4. Durability means resisting damage; portability means being easy to carry. An SSD has no moving read/write parts, so it handles shocks better. Size and weight also matter.

I Do / We Do pages · Guided practice

Q1. Input age; decision diamond age >= 18; adult/under 18 outputs; end.

Q2. IF age >= 18 THEN print('Adult') ELSE print('Under 18') ENDIF (valid equivalent accepted).

Q3. An IF chooses a branch; a FOR repeats instructions a known number of times.

You Do pages · Code First

```
fileGB = 12
freeGB = 10
print(fileGB - freeGB)
```

Expected result: 2

You Do pages · Independent questions

Q1. Diamond for a decision [1]; parallelogram for input/output [1] (other correct pairs accepted).

Q2. Output 7 [1]; first assignment stores 3 [1]; next adds 4 before print [1].

Q3. Selection tests length [1] and branches by condition [1]; iteration repeats the input prompt [1] until an acceptable value is entered [1].

Exit Ticket · Model answer

False

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: